

Week 5

Inner product and Length

Complex Matrices

Dot Product in $\mathbb{C}^n$

Properties

Norm and distance in $\mathbb{C}^n$

Complex Conjugate of Matrices

Hermitian Matrices

Properties

Orthogonal Matrices

Unitary Matrices

Properties

Diagonalization of Hermitian Matrices

Spectral Theorem

Schur's Theorem

- $i^2 = -1$

Inner product and Length

In $\mathbb{R}^n$

$$\|x\|^2 = x^T x = \overline{x}^T x$$

Complex Matrices

$$A + iB$$

Dot Product in $\mathbb{C}^n$

$$\langle x, y \rangle = \overline{x}^T y = \sum \overline{x_i} y_i$$

where $\overline{x_i}$ is a complex conjugate.

Properties

- $\langle x, y \rangle = \overline{\langle y, x \rangle}$
- $\langle \alpha x, y \rangle = \overline{\alpha} \langle x, y \rangle$

- $x \cdot \langle \alpha y \rangle = \alpha \langle x, y \rangle$
- $\langle x, x \rangle \geq 0$

Norm and distance in $\mathbb{C}^n$ =

$$\|u\| = (u \cdot u)^{1/2}$$

$$d(u, v) = \|u - v\|$$

Complex Conjugate of Matrices



Switch the signs on the imaginary terms

$$\text{Complex Conjugate of } A + iB = A - iB$$

Hermitian Matrices



Conjugate Transpose

$$M^H \text{ or } M^\dagger = M$$

Properties

- If A is a hermitian matrix, then all the eigenvalues of A are real.
- Hermitian matrices in a complex vector space are generalization of symmetric matrices in a real vector space.
- Hermitian matrices are orthogonally diagonalizable if their eigenvalues are distinct.
- The diagonal elements of a Hermitian matrix are real.
- If A is a hermitian matrix, then the eigenvectors corresponding to different corresponding eigenvalues are orthogonal.
- All real symmetric matrices are hermitian.

Orthogonal Matrices

Real matrices are orthogonal if

- columns form an orthonormal basis
- $A^T = A^{-1}$ or $AA^T = I$

Unitary Matrices

Complex matrices are unitary if

- columns form an orthonormal vectors
- $U^H = U^{-1}$
- $UU^H = U^H U = I$



A real unitary matrix is an orthogonal matrix.

Properties

- Eigenvalues of a unitary matrix have absolute value 1 i.e. $|\lambda| = 1$
- Eigenvectors corresponding to different eigenvalues of a unitary matrix are orthogonal.
- Length of a vector does not change when it is multiplied with the unitary matrix.

Diagonalization of Hermitian Matrices

- Every unitarily diagonalizable matrix is not a Hermitian matrix
- Every Hermitian matrix is unitarily diagonalizable.

Spectral Theorem



Any Hermitian matrix is unitarily diagonalizable.

For a real symmetric matrix A , we can find an orthogonal matrix Q such that

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$$A = Q \Lambda Q^T$$

Schur's Theorem

Any $n \times n$ matrix A is similar to an upper triangular matrix T i.e. there exists a unitary matrix U such that

$$A = UTU^\dagger$$